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Capsules with a hierarchical shell structure assembled by aminoglycosides and DNA *via* the kinetic path†

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Aminoglycosides are capable of expelling water molecules when forming a complex with DNA *via* electrostatic interaction. The "water-proof" nature of the complex leads to the formation of capsules, which possess hierarchical shell structures with a smooth and rigid outer layer and a viscoelastic inner layer.

The ability of DNA to self-assemble into diverse nanostructures creates abundant opportunities for material design and synthesis.[1–3] One widely explored route of DNA self-assembly is through electrostatic complexation of DNA,[4–7] being negatively charged, with cationic molecules in aqueous solution. The entropic gain from the concurrent release of counterions is known to drive such electrostatic complexation.[8] The electrostatics-driven complexation of DNA also plays important roles in biology, with notable examples including histone-mediated DNA packaging of eukaryotic chromatin[9,10] and protamine-induced DNA condensation[11] in sperm heads. Inspired by DNA condensation *in vivo*, a variety of positively charged reagents, such as synthetic polymers, lipids and liposomes, peptides and proteins, and multivalent cations, have been investigated to probe the governing principles of complexation and to synthesize new advanced materials for applications in nanotechnology and medicine.[12–15]

The resultant complexes of DNA and cationic molecules exhibit wide-ranging nanostructures and multi-functionalities conducive to future applications. Moreover, their structures can be modulated by a large number of factors, such as the charge ratio, mixing order, salt concentration, and solvent quality.[16–19] One factor of particular importance is kinetics, which is observed to not only regulate the complexation process but also lead to novel non-equilibrium structures. Kinetic trapping thus offers a strategy to fabricate desirable DNA structures. However, the kinetically trapped DNA structure generally evolves in aqueous solution with time due to the relaxation and redistribution of the charged molecules.[20,21] To circumvent such undesirable changes, one common practical approach is to completely remove water (*i.e.*, drying) right after the complexation so as to "lock" the DNA structures into designed shape or morphology.[22,23]

As the structural relaxation mainly arises from the mobility of small molecules within DNA nanostructures, the depletion of water molecules around DNA is the key to "lock" the DNA structure into a metastable state. Alternative to the "drying" method, it has been reported that polyols, such as glycerols, possess the ability to replace the water molecules in DNA hydration sites and modify the electrostatic interactions between the DNA backbone and its surrounding atmosphere of counterions.[24] Therefore, cationic reagents with multiple hydroxyl groups should hold promise to form kinetically stable structures with DNA in aqueous solution, which has not yet been demonstrated to our knowledge.

Herein, we chose kanamycin, an amino modified sugar widely used as antibiotics, and studied its interaction with genomic double-stranded DNA (from salmon testes) in aqueous solution. Kanamycin contains four positive charges and seven hydroxyl groups. It can quickly interact with DNA upon mixing due to the small size or fast diffusion rate, physically cross-linking DNA strands *in situ* in aqueous solution. The morphology of the complex depends on how DNA and kanamycin are mixed together. In this work, DNA solution is added into kanamycin solution, so that the fast moving kanamycin molecules are able to quickly "lock" the DNA droplet into a fixed shape and "seal" the freely moving DNA molecules inside. Small molecules and nanoparticles are also included in DNA solution beforehand to probe the locking and sealing capacity of kanamycin.

We firstly studied the interaction capability of kanamycin and long-chain genomic DNA in dilute solutions by laser light

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† Electronic supplementary information (ESI) available: Experimental section, differential sedimentation assay, polarized optical microscopy (POM), SEM images, and DLS results. See DOI: 10.1039/c4cc03508a

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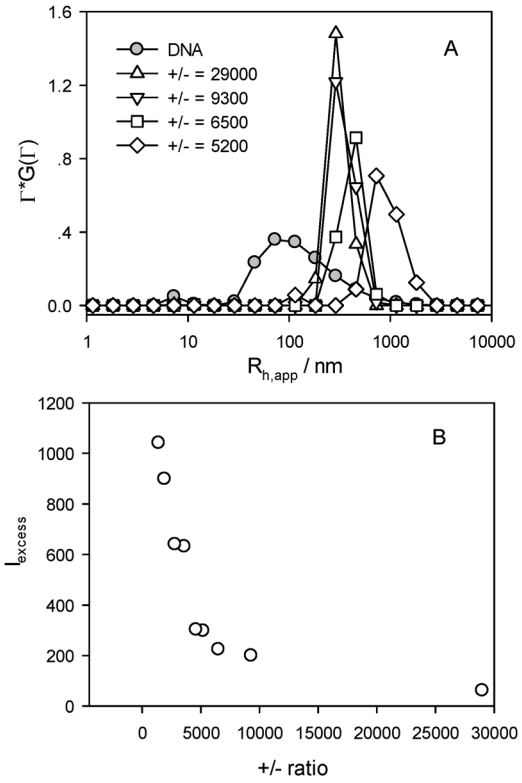


Fig. 1 Selected size distributions (A) and I_{excess} at 30° (B) of DNA–kanamycin complexes at different +/− charge ratios. Stock concentrations: c(kanamycin) = 1.0 × 10^{-2} g mL^{-1} and c(DNA) = 1.0 × 10^{-4} g mL^{-1}.

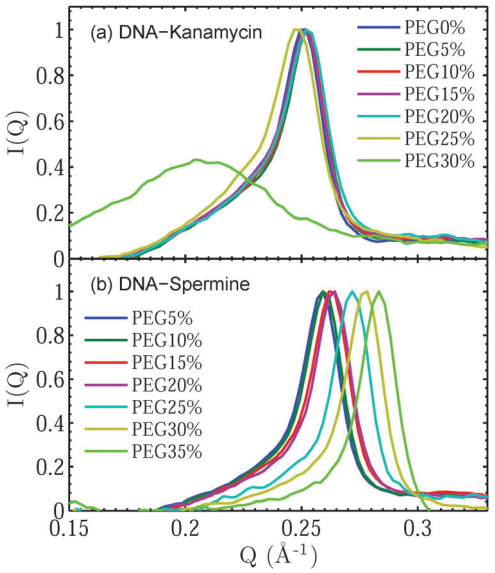


Fig. 2 X-ray diffraction peaks of kanamycin– and spermine–DNA complexes in the presence of PEG (molecular mass of 8000 Dalton) at varied concentrations (weight/weight). Here Q is the magnitude of the scattering vector and is defined as $4\pi\sin\theta/\lambda$, where θ is half of the scattering angle and λ is the X-ray wavelength. (a) Kanamycin–DNA complex. Note that a sudden jump to larger DNA spacing was observed at 30% PEG, the origin of which remains to be resolved. (b) Spermine–DNA complex.

scattering. Fig. 1 shows the size distribution and the excess scattered intensity at 30° as DNA is added to kanamycin solution. Data on the original DNA stock solution are also included for better comparison. Both the size and the excess scattered intensity sharply increase with the addition of DNA. With one small drop of DNA added into kanamycin (+/− charge ratio of 2.9 × 10^4), the average hydrodynamic radius ($R_{h,app}$, "app" represents apparent) immediately increases from 90 nm (the size of DNA itself) to 290 nm; and the excess scattered intensity also increases by a factor of 40, indicating the fierce interaction between DNA and kanamycin. The contribution of free DNA to the scattered intensity is not prominent because its size is small. At a +/− ratio of about 1.0 × 10^3, far from unity, the solution becomes turbid and precipitation is visible to naked eyes. Unlike the interaction between DNA and other multivalent cations such as tetravalent spermine or trivalent spermidine,[25–27] no reentrant condensation is observed at higher concentration of kanamycin. This is further corroborated by an independent differential sedimentation assay showing the persistence of DNA precipitations at all kanamycin concentrations above 1 mM (Fig. S1, ESI†). The extraordinary complexation behaviour is somewhat surprising given that kanamycin is bulkier than spermine or spermidine and expected to be less effective in condensing DNA. We speculate that this might originate from the hydrogen bonding interaction between kanamycin hydroxyl groups and DNA, as well as the dehydration of DNA backbone.

To elucidate the nature of kanamycin–DNA interactions, we next applied polarized optical microscopy and X-ray diffraction (XRD) to probe the microscopic structures of the resultant complexes. Liquid crystalline ordering is evidenced by the strong optical birefringence (Fig. S2, ESI†), indicative of side-by-side alignment of DNA strands. The high degree of DNA ordering is established by the well-defined XRD peaks (Fig. 2a), the widths of which are about the same as that from DNA–spermine complexes (Fig. 2b), noting that spermine also has four positive charges but lacks hydroxyl groups. XRD can further give the spacing between DNA strands from its peak position. The DNA–DNA inter-axial spacing in DNA–kanamycin complexes is determined to be ~29.5 Å in 10 mM kanamycin (Fig. S3, ESI†). Compared with the spacing in DNA–spermine complexes, it is about 1 Å larger, which is probably due to kanamycin's bulkier size. In order to examine whether the hydroxyl groups of kanamycin can replace water molecules around DNA as reported for polyols,[24] the osmotic stress method (OSM) was then applied to probe the extent of hydration in the kanamycin–DNA complexes.[28] In this context, OSM explores the use of neutral osmolytes, such as poly(ethylene glycol) (PEG), to exert forces on molecular assembles, whose physical origin can be understood in terms of molecular crowding or depletion force. As a result, this osmotic force "pushes out" interstitial water molecules if present and compresses the macromolecular assembly, which can be monitored by the shifts in XRD peak positions. Fig. 2 shows the XRD peaks of kanamycin– and spermine–DNA complexes in the presence of PEG at varied concentrations. The trends observed from the two complexes are in sharp contrast. For spermine–DNA, its

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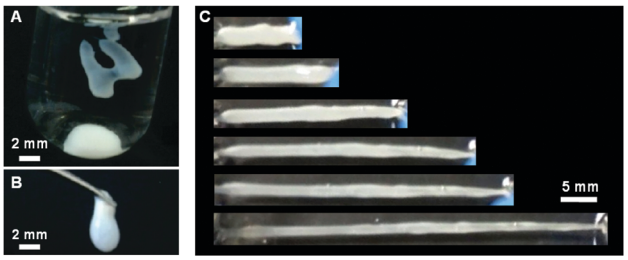


Fig. 3 (A & B) DNA–kanamycin capsules of various morphologies. (C) Shape elongations after stretching. c(DNA) = 1.0 × 10^{-2} g mL^{-1}; c(kanamycin) = 1.0 × 10^{-1} g mL^{-1}.

DNA–DNA spacing decreases as the PEG concentration increases, evident from the shifts of XRD peaks to higher Q positions (Fig. 2b). This is consistent with the ''pushing-out'' of interstitial water molecules in spermine–DNA complexes. However, for kanamycin–DNA, its DNA–DNA spacing remains nearly constant and in fact slightly increases as PEG concentration increases (Fig. S3, ESI†). Therefore, distinct hydration states are observed from kanamycin– and spermine–DNA complexes and, remarkably, there appears to be no or minimal water in kanamycin–DNA complexes. The hydroxyl groups of kanamycin thus result in water-tight DNA complexes, likely due to their ability to act as water substitutes.

Given the tight packing and highly water-proof nature of the DNA–kanamycin complex, we further prepared capsules of various shapes based on the assembly of kanamycin onto DNA by immersing DNA solution (above the overlap concentration) into kanamycin solutions. Fig. 3 shows various shapes of capsules, such as tube-like, spherical, and H-shaped. The morphology of the final capsules depends on the shape of the fluid DNA injected. We can prepare spherical capsules by directly pipetting a droplet, or membrane-like capsules by squeezing DNA solution through a narrow slit. We can easily move the spherical capsules in solution using tweezers (Fig. 3B). The capsules are stable in aqueous solution up to days and possess relatively high mechanical strength. For example, we can stretch the tube structure using tweezers until it reaches both ends of the dish (Fig. 3C and Movie S1, ESI†). The tube length increases by a factor of at least 4. The elastic ability of the capsule provides an easy way for further fabrication and application.

The SEM experiment was conducted to reveal the structure of the capsule at the microscopic level. The whole capsule is freeze-dried, and part of the surface is removed using tweezers to expose the inside of the capsule. As shown in Fig. 4A, a shell with a hierarchical structure is formed by kanamycin and DNA. The outermost surface of the shell is smooth and rigid (Fig. 4B), which is attributed to the close packing of kanamycin molecules (see Fig. S4a, ESI†). The wrinkles and cracks on the surfaces are formed during the freeze-drying process. Further inside of the shell is the network of thick fibers (Fig. 4C). The thickness of fibers decreases radially, suggesting that the formation of the network is related to the diffusion of kanamycin. The pore size of the network ranges from several to tens of microns. In the center

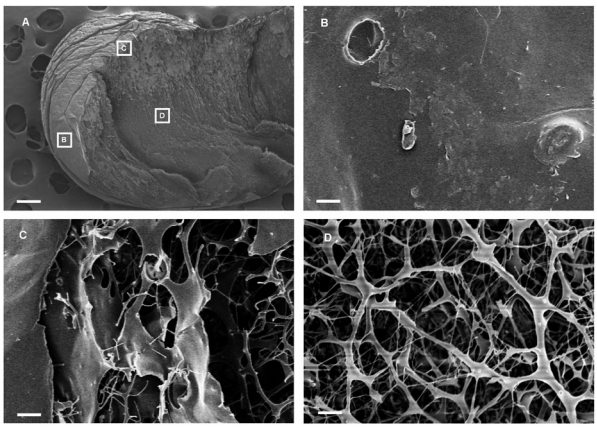


Fig. 4 SEM images of DNA–kanamycin capsules at selected regions. The images in (B–D) were taken in the areas indicated in the white boxes in (A). c(DNA) = 1.0 × 10^{-2} g mL^{-1}; c(kanamycin) = 1.0 × 10^{-1} g mL^{-1}. The scale bar is 200 μm in (A), but 4 μm in (B–D).

of the capsule is the network of the thin fibers (Fig. 4D). This network likely contains pure DNA as the concentration is above the overlap concentration.

On the basis of the above results, the mechanism of capsule formation can be understood as follows. When a droplet of DNA solution is added into kanamycin, the fast moving kanamycin molecules quickly interact with the DNA at the outer layer of the droplet. Since each kanamycin has four positive charges and can interact with more than one DNA chain, it serves as a cross-linking point. Moreover, the special ''water-proof'' or the dehydration of the DNA–kanamycin complex enhances the toughness of the formed shell. At DNA concentrations above the overlap concentration, the formation of the hard shell around the droplet is faster than the diffusion of DNA molecules, locking the DNA molecules inside the capsule. Therefore, the rigidity of the shell maintains the shape of the droplets, and the physical cross-linking networks render the capsule certain elasticity. As such, kanamycin is different from other carbohydrate-based polycations or chitosan[29,30] in the formation of a complex with DNA. Chitosan and its derivatives, due to their large size and lacking the ability of ordered packing, exhibit a complexation behavior similar to the regular polycations.

Fluorescence probes are used to test the sealing ability of the capsule formed by kanamycin and DNA. Since the diffusion of molecules plays a role in the preparation of capsules, we chose two fluorescence probes, rhodamine B and rhodamine B-labeled liposome. Rhodamine B is a small molecule and carries one positive charge. The size of the rhodamine B-labeled liposome is 100 nm in diameter (Fig. S5, ESI†), comparable to the size of DNA. Both rhodamine B and rhodamine B-labeled liposome have no prominent interactions with either DNA or kanamycin. The release of the probes from DNA solution (no capsule) is used as the control. As shown in Fig. 5A, it takes about 15 min for rhodamine B to diffuse out of the DNA solution. However, the release rate is significantly reduced when forming a DNA–kanamycin capsule (Fig. 5A). The burst release of 60% at the

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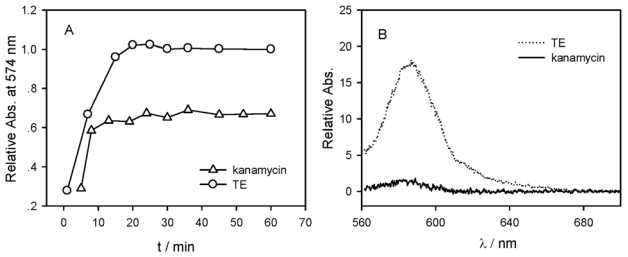


Fig. 5 Encapsulation of small molecule rhodamine B (A) and nanoparticle rhodamine B-labeled -liposome (B) in DNA–kanamycin capsules. (A) The relative intensity is normalized to the asymptotic intensity of rhodamine B. c(DNA) = 1.0 × 10^{-2} g mL^{-1}; c(kanamycin) = 1.0 × 10^{-1} g mL^{-1}, 1 × TE buffer (pH = 7.5, 10 mM Tris-base, 1.0 mM EDTA).

beginning is caused by the diffusion of rhodamine B during the formation of the capsule. No burst release is observed when rhodamine B-labeled liposome is used as the probe. Nearly all the liposomes are trapped in the capsule, as the emission peak of rhodamine B cannot be detected outside the capsule (Fig. 5B). The SEM image clearly shows that the liposomes are trapped inside the fibers formed by DNA (Fig. S4b, ESI†). All these results demonstrate that the shell of the capsule is effective in preventing the content, either small or large molecules, from leaking out of the capsule.

Aminoglycosides, traditionally used as antibiotics, possess both polyols and charged groups that are conducive to molecular self-assembly. Using kanamycin as a model, we investigated the complexation of aminoglycosides with double-stranded DNA by a variety of physical techniques. While the complexation is ascertained to be driven by electrostatics, results show that kanamycin is able to expel the hydration water from DNA molecules upon forming a complex. By taking advantage of the unique interactions, we successfully prepared a novel class of capsules made up of DNA and aminoglycosides. The morphology of the capsule is determined by the shape of the DNA droplet before being immersed in kanamycin solution; no chemical reaction or complex formulation is needed. The capsule contains a shell with a hierarchical structure. The smooth and rigid layer maintains the shape and the physically cross-linked network renders the capsule elasticity. Such a shell is able to effectively seal the molecules, either small or large, inside the capsule. Since both DNA and kanamycin are biomolecules derived from life forms, such capsules have high biocompatibility. The capsule with a hierarchical shell structure should be useful in wound dressing and drug delivery.

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Supporting Information for

Capsules with Hierarchical Shell Structure Assembled by Aminoglycoside and DNA via Kinetic Path

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Experimental Section

Materials: Kanamycin sulfate, salmon DNA (~2000 bp), Rhodamine B, and 1,2-dipalmitoyl-sn-glycero-3-phosphatidylcholine (DPPC, purity > 99%), were purchased from Sigma-Aldrich (St Louis, US). Rhodamine B-labeled lipid, 1,2-dipalmitoyl-sn-glycero-3-phosphoethanolamine-N-(lissamine rhodamine B sulfonyl) (Rh-DPPE, purity > 99%), was purchased from Avanti Polar Lipids, Inc (AL, US). Whatman Nuclepore Track-Etch membrane for extrusion was purchased form GE Healthcare (Germany). Rhodamine B-labeled liposome (DPPC: Rh-DPPE = 99.9: 0.1, molar ratio) was prepared by the extrusion method[1] with 100 nm membrane.

Fabrication of DNA-kanamycin microcapsules: The capsule can be prepared into a variety of shapes either by changing the injection pattern or using the pushing and dragging method. Using the spherical capsule as an example, the DNA solution at c(DNA) = 1.0×10^{-2} g/mL to c(DNA) = 5.0×10^{-2} g/mL was added into kanamycin solution (c (kanamycin) = 1.0×10^{-1} g/mL) drop by drop. The size of the droplet can be tuned by changing the diameter

of the outlet. The droplet of DNA solution was transparent. However, a white and thin shell was quickly formed around the entire droplet, resulting in a microcapsule. The shell matured in about 1-10 min, depending on the concentration of DNA and kanamycin. The smallest diameter of the capsule was in the order of few hundred micron. The shell of the capsule was flexible in solution. It can be stretched with ease. However, the capsule would shrink, become brittle, and dry after being pulled out from water.

Microstructure characterization: The interaction of kanamycin and DNA at dilute solution was investigated by laser light scattering (LLS) and X-ray diffraction (XRD) via the osmotic stress method. A commercialized spectrometer BI-200SM Goniometer was used to perform both static light scattering (SLS) and dynamic light scattering (DLS), over a scattering angular range of 20-120°. Detailed LLS procedures were described in our previous papers.[2] Samples for LLS were passed through 0.45 μm syringe filter to remove dust. XRD experiments were performed with an in-house XRD setup (3×10^7 photons/second, beam size of 0.8×0.8 mm) where radiation damage is negligible. To investigate the interface of the microcapsule, freeze-dried SEM samples (microcapsules formed by adding DNA (c(DNA) = 1.0×10^{-2} g/mL) to kanamycin (c (kanamycin) = 1.0×10^{-1} g/mL)) were stained by Au and examined using Hitachi-S4800 operated at 3 kV.

Encapsulation of molecules and nanoparticles: The mixture of DNA (c(DNA) = 1.0×10^{-2} g/mL) co-dissolved with small molecules Rhodamine B or Rhodamine B-labeled liposome solutions were added into kanamycin solution with a pipette. The small molecules or nanoparticles were encapsulated into the formed capsule. The fluorescent spectra were recorded at room temperature using Hitachi-F7000. DNA co-dissolved with Rhodamine B or Rhodamine B-labeled liposome added to TE buffer was taken as references. The total Rhodamine B intensity was recorded after dissolving the capsule with 2 M NaCl.

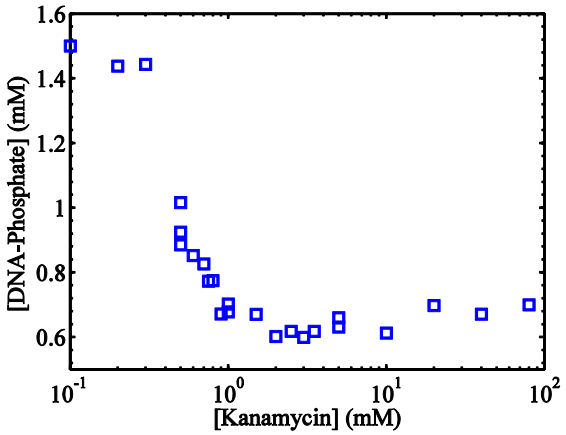


Figure S1. Differential sedimentation assay of kanamycin-induced DNA condensation. DNA and kanamycin stock solutions are mixed at appropriate volumes to give a sample series with constant DNA concentration (1.5 mM DNA-Phosphate) and varied kanamycin concentration. Each sample is vortexed and then centrifuged at 14000×rpm for 15 minutes. The concentration of DNA-Phosphate in the supernatant is shown as a function of kanamycin concentration . 1×TE buffer (10 mM Tris, 1 mM EDTA, pH 7.5) is used in all samples.

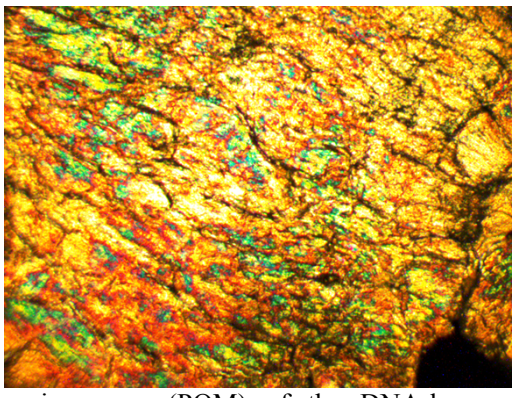


Figure S2. Polarized optical microscopy (POM) of the DNA-kanamycin complex. Strong optical birefringence is observed indicative of liquid crystalline ordering.

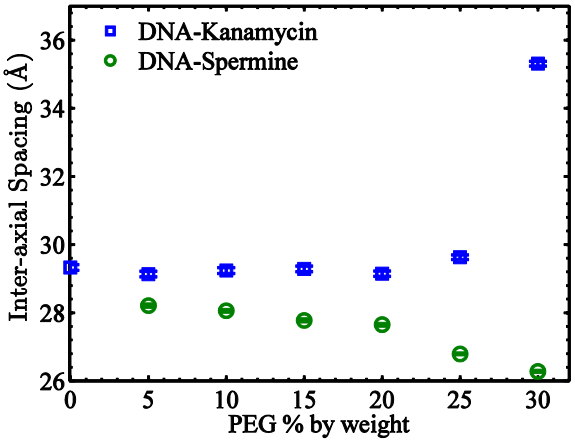


Figure S3. DNA-DNA spacings of the kanamycin- and spermine-DNA complexes as a function of the PEG concentration (weight/weight). Qualitatively different trends are observed.

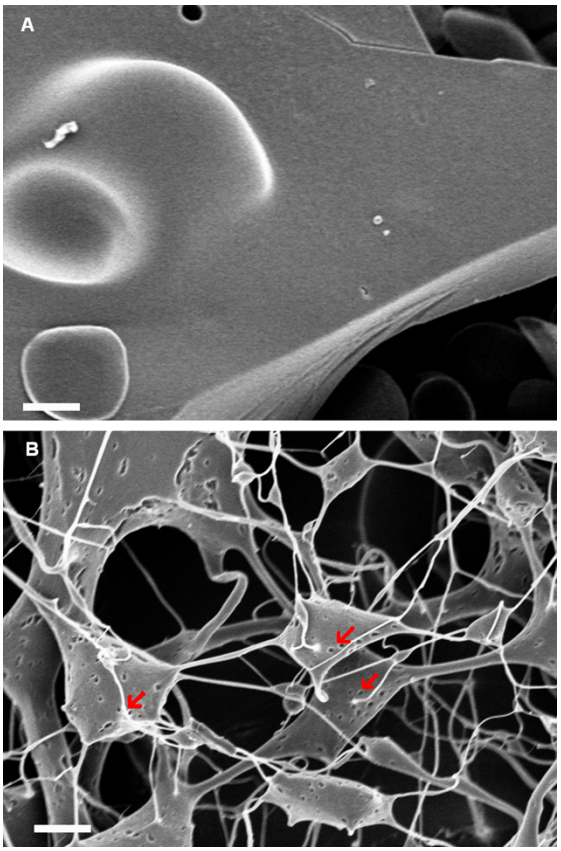


Figure S4. SEM images of kanamycin (A) and liposome-embedded capsule (B). Scale bars are 1μm. The red arrows in (B) indicate the location of liposomes.

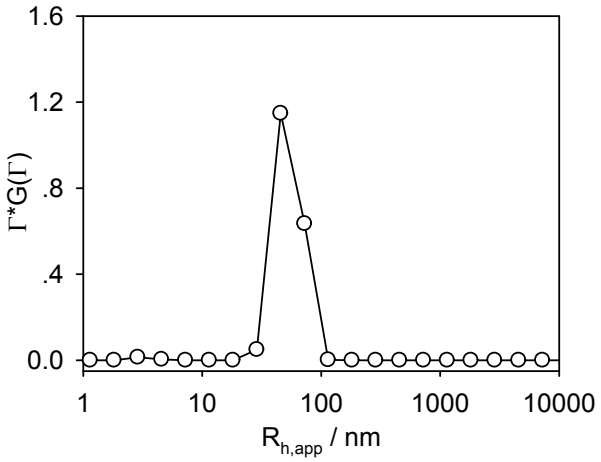


Figure S5. The size distribution of Rhodamine B-labeled liposome at 30°. c(liposome) = 5.0 nM.